

Prediction Is Not Identification

A minimal benchmark for choosing the experiment that matters

Jawaun Brown Human director; experiment code and drafts generated under review 2026-07-27 · IDENT v1

Abstract

Observational prediction can be perfect while mechanism identification remains impossible. We introduce **IDENT**, a one-shot benchmark where two or more hidden mechanisms are indistinguishable under a supplied observation set, yet at least one permitted intervention separates them. We prove a finite passive-identification impossibility bound, ship a deterministic generator with exact separator annotations across three symbolic domains, and evaluate trivial baselines plus frontier models under a frozen one-intervention protocol. Construction gates pass. An oracle weakest identifying separator reaches 100% final identification. Strong models usually intervene rather than bluff, but final identification remains far below oracle—even when separator choice is perfect—revealing a residual failure in update-after-evidence.

1. Claim and scope

Operational claim. There exist tasks on which a passive learner cannot exceed chance at naming the true mechanism, while a single cheap intervention identifies it exactly.

IDENT is intentionally narrow. It is not a general theory of intelligence, a transformer training run, or a natural-language science suite. The contribution succeeds only if three pieces land together:

1. a formal impossibility result for passive learners on the benchmark family;
2. a compact generator with known latent mechanisms and exact separators;
3. an evaluation showing whether models recognize underdetermination and choose (then use) the right experiment.

2. Minimal theory

2.1 Objects

- Finite hypothesis set $(H = \{h_1, \dots, h_k\})$.
- Experiment set (G) with costs $(c(g) \geq 0)$.
- Response function $(R(h, g))$.
- Initial passive family $(G_0 \subset G)$.

2.2 Experiment-relative equivalence

For $(A \subseteq G)$,

$[h_i \sim_A h_j \text{ iff } R(h_i, g) = R(h_j, g) \text{ for all } g \in A.]$

The live class $(S = [h^\star]_{\sim_{G_0}})$ is the set of mechanisms still compatible with all passive observations.

2.3 Weakest identifying separator

Intervention (g) **separates** (S) when outcomes $(\{R(h, g) : h \in S\})$ are not unique. It **identifies** $(h^\star)$ when observing $(R(h^\star, g))$ leaves a singleton live class. IDENT annotates minimum-cost identifying separators.

2.4 Passive identification impossibility

Theorem (benchmark form). If $|S| = m \geq 2$ and the held-out label is non-constant on (S) , then no learner receiving only the passive transcript can exceed worst-case accuracy $(1/m)$ at recovering the true hypothesis.

Proof sketch. Any two members of (S) induce identical passive transcripts. A deterministic map returns one answer on that transcript and is therefore wrong on at least one world. A randomized map has minimax accuracy at most $(1/m)$ against an adversary choosing the true member of (S) . $\square$

Active sufficiency. IDENT v1 generates only items with at least one one-step identifying separator, so failure is not caused by an impossible task.

3. Benchmark

3.1 Task interface

Each item supplies prior observations, a hypothesis menu, and candidate interventions with costs. The model may answer immediately or spend exactly one intervention, receive the outcome, then answer.

3.2 Domains

Domain	Latent object	Passive masking
boolean_causal	two-input Boolean mechanisms	agree on an observed input support
finite_state	tiny acceptors over $\{0,1\}^{\leq 3}$	agree on observed traces
small_programs	integer functions	agree on supplied test inputs

Every item includes a wasteful “more of the same” distractor that cannot separate the live class.

3.3 Dataset

Deterministic seed 20260727 yields 1000 items: train 700 / dev 150 / test 150. Validators enforce live-class size (≥ 2) , existence of an identifying separator, exact minimum-separator annotations, passive chance bounds, and basic leak checks.

3.4 Metrics

Separator accuracy, weakness regret, false-certainty rate, post-intervention identification, final accuracy, and efficiency (identification gain / cost).

4. Pre-registered gates

Gate	Result
G1 Formal ambiguity	PASS
G2 Separability / active sufficiency	PASS
G3 Passive bound (answer-now $\leq$ chance + Hoeffding slack)	PASS
G4 Oracle solvability ($\geq 99\%$ final ID)	PASS (100%)
G5 Capability gap vs oracle	PASS
G6 EIG nontriviality	PASS
G7 Label/order robustness	PASS

5. Results

5.1 Baselines (full test, n=150)

Baseline	Separator	False certainty	Final acc
answer_now	0.00	1.00	0.50

random_intervention	0.55	0.00	0.69
max_output_variance	1.00	0.00	0.92
expected_information_gain	1.00	0.00	0.93
oracle_weakest_separator	1.00	0.00	1.00

Answer-now is maximally falsely certain and remains near chance. Exact information gain already selects separators on this symbolic menu; IDENT is therefore a measurement instrument, not a claim of a new IG principle.

5.2 Frontier models via OpenRouter (test slice, n=40)

Frozen protocol; temperature 0; one permitted intervention.

System	Separator	False certainty	Post-intervention ID	Final acc
oracle	1.00	0.00	1.00	1.00
anthropic/claude-sonnet-4	1.00	0.05	0.58	0.55
openai/gpt-4o-mini	0.74	0.03	0.41	0.45
google/gemini-2.5-flash	0.80	0.00	0.25	0.23

G7 reshuffle (n=15) leaves separator accuracy within 0.07 for Claude and GPT-4o-mini.

5.3 Failure shape

Models usually intervene. The dominant residual is **update-after-evidence**: Claude matches oracle separator accuracy yet reaches only 0.55 final identification. Weaker models also choose non-separating or invalid menu actions. This matches the plan’s interpretation that the capability gap can lie in experiment use and hypothesis bookkeeping, not only in refusing to act.

6. What this does not show

- That experiment-relative equivalence is a new concept.
- That information gain is a new learning principle.
- That model failure proves absence of “understanding.”
- That the result transports to natural-language science wrappers without further leak tests.

7. Reproduce

```
python3 -m experiments.ident.experiment
python3 -m pytest tests/test_ident.py -q
doppler run --project cofounder --config dev -- \
  python3 -m experiments.ident.eval.run_models --limit 40 --no-robustness
```

Package: experiments/ident/. Paper source: papers/ident/paper.md.

8. Conclusion

IDENT makes one missing capability hard to ignore: sounding certain—or even requesting an experiment—is not the same as identifying a mechanism. Passive prediction is provably bounded on the live equivalence class; the correct one-step separator is exactly known; and current strong models leave a reproducible gap between intervention and identification.